

Tristan Anglin

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Portfolio: tristananglin.github.io

Summary

Systems-focused Software Engineer & Software Developer with a strong foundation in Object-Oriented Programming (OOP), C++, C#, and core Computer Science principles. Proven expertise in Full-Life Cycle Software Development (SDLC), low-level custom graphics pipelines, client-server architectures, and Asynchronous Programming. Combines hands-on experience building scalable applications, managing memory allocation, and debugging complex logic with an efficient, AI-accelerated workflow. Strong background in Agile/Scrum environments, Version Control (Git/GitHub), and cross-functional technical leadership.

Technical Skills

- **Programming Languages:** C++, C#, JavaScript, TypeScript, Python, Java, SQL, HTML/CSS.
- **Software Engineering & Architecture:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Asynchronous Systems, Client-Server Networking, State Replication, Event-Driven Architecture, Memory Management, Performance Optimization.
- **Frameworks & Developer Tools:** Unreal Engine (C++ & UMG), Unity, Custom OpenGL Pipelines, Visual Studio, Eclipse, CMake, Git, GitHub, Jira, Node.js, React, Tailwind CSS.
- **Methodologies & Workflows:** Agile/Scrum, Full-Stack Development, CI/CD Workflows, Code Reviews, Unit Testing, Debugging, Hardware Abstraction, and AI-Assisted Development (Code Diagnostics & Refactoring).

Relevant Education

Algonquin College | Ottawa, ON — *Advanced Diploma in Game Development*

Sept 2023 – April 2026

- **Academic Excellence:** Cumulative 3.86 GPA, consistent Dean's List Honor Roll standing. Focused heavily on core Computer Science, advanced mathematics, data structures, and software architecture.

Relevant Experience

Algonquin College, Ottawa, ON — Lead Systems & Software Developer | *Blood & Lineage*

Sept 2025 – April 2026

- Architected and implemented modular, event-driven user interface components and software modules using C++ and component-based frameworks, managing complex real-time data binding for multi-client applications.
- Designed and maintained high-performance UI rendering pipelines and asynchronous layout updates, guaranteeing sub-millisecond frame rendering and strict Performance Optimization.

- Engineered core Client-Server Networking logic to securely handle session hosting, socket management, and asynchronous State Replication across distributed instances.
- Directed the technical roadmap, Version Control strategies, and build stability using Git/GitHub and Jira in an Agile/Scrum team environment consisting of 11 engineers.

**Systems Developer | Custom 2D Graphics Engine & Strategy Game (Solo) | Ottawa, ON
Dec 2024**

- Engineered a lightweight 2D rendering engine from scratch utilizing Modern C++ and OpenGL, achieving optimized low-level Hardware Abstraction, custom shader pipeline management, and efficient heap memory allocations.
- Implemented complex graph algorithms (*A Pathfinding**) and spatial data structures to execute real-time collision detection, dynamic routing updates, and rigorous state validation logic.
- Developed a custom camera viewport controller featuring a Top-Down Depth Sorting (Y-sorting) algorithm, minimizing draw calls and optimizing CPU instruction cycles during intensive vector rendering transformations.

**Peer Tutor & Program Assistant | Algonquin College, Ottawa, ON
Sept 2024 – April 2026**

- Mentored software development students in diagnosing complex logic errors, memory leaks, concurrency issues, and performance bottlenecks in C++ and C# codebases.
- Conducted rigorous Peer Code Reviews focused on clean code architecture, code maintainability, and industry-standard design patterns.
- Synthesized abstract Systems Architecture and software engineering principles into digestible, actionable debugging and troubleshooting guidelines.

Technical Volunteer | *Girls Who Game* — Dell Technologies | 2024 – Present

- Evaluated student coding projects and provided constructive feedback to foster STEM growth.
- Streamed virtual Esports tournaments, managing live technical execution and troubleshooting in the streaming app space.

References

Konstantin Titov | Coordinator/Professor, Game Development Program 613 727 4723 Ext 2412 Algonquin College | 1385 Woodroffe Avenue | Ottawa | Ontario | K2G 1V8 | Canada

- Konstantin has overseen my development since 2023, observing my progress from an entry-level student to a technical leader. He can speak to my 3.86 GPA and my performance as a Peer Tutor and Program Assistant, where I provide technical mentorship to my peers.

Dushan Horvat | 3rd year Coordinator/ Professor, Algonquin College | 1385 Woodroffe Avenue | Room J315B | Ottawa | Ontario | K2G 1V8 | Canada

- Dushan has been a key mentor during my final year, providing critical guidance on game scope and project management. His oversight was instrumental in helping me navigate the complexities of full-cycle development, ensuring my projects remained both ambitious and achievable.

Ryan McDonald

Former Supervisor | RONA | 613-406-2125 | RyanMcDonald@gmail.com

- Ryan was my direct supervisor during my time at RONA, where I developed my skills in team collaboration and client-facing communication.